Theoretical Foundations for Speed and Torque Control of a Switched Reluctance Motor

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***Abstract*—Switched reluctance motors (SRMs) are attractive for electric-drive applications due to robust construction, absence of rotor magnets, and wide speed capability. Their nonlinear inductance profile and commutation behavior, however, make high-performance torque and speed control challenging. This paper consolidates the theoretical background for an SRM speed/torque control project milestone. We summarize machine fundamentals, nonlinear dynamic equations, converter operation, and established control methods including hysteresis current control, fixed-frequency PWM, turn-on/turn-off angle scheduling, and torque sharing functions (TSF). A cascaded control architecture is presented as a practical implementation template for subsequent MATLAB/Simulink realization.**

***Index Terms*—Switched reluctance motor, electric drives, torque ripple, torque sharing function, PI speed control**

I. Introduction

SRMs are increasingly considered for applications that prioritize reliability, fault tolerance, and reduced dependence on rare-earth materials [1], [2]. The machine structure is simple—a salient-pole stator carrying concentrated windings and a salient-pole rotor with neither magnets nor windings—but control complexity is high because torque is generated through position-dependent inductance rather than sinusoidal back-EMF. As a result, both commutation logic and current regulation must be tightly synchronized with rotor position [3].

Early analytical treatments demonstrated that the SRM's nonlinearity is amenable to feedback-linearizing control [4], while subsequent work focused on practical converter topologies, current regulation, and torque-ripple suppression. This milestone addresses the theoretical layer that precedes implementation. The objective is to establish a coherent model-to-control chain—from machine equations through converter switching to torque sharing and speed regulation—that can be translated into MATLAB/Simulink simulation and, subsequently, real-time drive experiments.

II. SRM Construction and Inductance Profile

A common SRM configuration is the three-phase 6/4 geometry: six stator poles each carrying a concentrated coil, and four rotor poles with no windings or magnets. Diametrically opposite stator coils are connected in series to form one phase, yielding three independent phases (A, B, C) displaced by 30° electrical [1].

The phase inductance $L(\theta)$ varies periodically with rotor position between two extremes. At the *aligned* position (θ_a), a rotor pole is fully aligned with an energized stator pole pair, the magnetic reluctance of the airgap path is minimized, and inductance reaches its maximum $L_{\max}$. At the *unaligned* position (θ_u), the rotor interpolar axis faces the stator pole, reluctance is maximized, and inductance has its minimum $L_{\min}$.

Positive motoring torque is produced only in the rising-inductance interval ($dL/d\theta > 0$), hence each phase must be energized during that region and demagnetized before the falling-inductance zone. The ratio $L_{\max}/L_{\min}$ directly influences torque capability; values of 6–10 are typical for well-designed machines [2].

Under magnetic saturation the inductance becomes a function of both θ and i, i.e. $L(\theta, i)$. A nonlinear look-up table, typically obtained from finite-element analysis (FEA), is therefore required for accurate simulation.

III. SRM Dynamic Model

A. Electrical and Magnetic Relations

For phase voltage v_k, current i_k, resistance R, and flux linkage ψ_k, the standard phase relation is

$$v_k = Ri_k + \frac{d\psi_k}{dt}. \tag{1}$$

With position-dependent inductance, this expands to

$$v_k = Ri_k + L(\theta, i_k)\frac{di_k}{dt} + i_k\frac{\partial L}{\partial \theta}\omega, \tag{2}$$

where ω is rotor speed. The last term represents motional EMF and grows with speed, eventually preventing the controller from maintaining the desired current waveform above a characteristic *base speed*.

Under linear (unsaturated) assumptions, flux linkage reduces to $\psi_k(\theta, i_k) = L(\theta)\, i_k$. For practical SRMs, saturation requires the nonlinear map $\psi(\theta, i)$, commonly implemented as two-dimensional lookup tables derived from FEA or measurements [2].

B. Torque and Mechanical Dynamics

The phase torque expression under the linear inductance assumption is

$$T_{e,k} = \frac{1}{2}\, i_k^2 \frac{dL_k(\theta)}{d\theta}, \tag{3}$$

and total electromagnetic torque is $T_e = \sum_k T_{e,k}$. In the saturated regime, torque is more accurately obtained from the co-energy integral. Mechanical motion obeys

$$J\frac{d\omega}{dt} = T_e - T_L - B\omega, \tag{4}$$

where J is rotor inertia, T_L the load torque, and B a viscous damping coefficient.

IV. Converter Topology

The asymmetric half-bridge converter remains the dominant SRM topology because each phase can be energized and demagnetized independently using two switches and two diodes [1]. Three switching states are available per phase, summarized in Table I.

TABLE I
Asymmetric Half-Bridge Switching States

Mode	Q1	Q2	Phase voltage
Magnetisation	ON	ON	$+V_{dc}$
Freewheeling	ON	OFF	≈ 0
Demagnetisation	OFF	OFF	$-V_{dc}$

Fast current decay during demagnetisation is essential: applying the full negative DC-link voltage forces the current to zero quickly, preventing negative torque in the falling-inductance region. The speed at which demagnetisation completes limits the motor's maximum operating speed and determines the required turn-off angle advance [3].

V. Current Control Strategies

A. Hysteresis Current Control

At low-to-medium speed the back-EMF is small and current must be actively limited. Hysteresis (bang-bang) control compares the reference current i^* with the measured phase current using a hysteresis band of width Δi:

- $i < i^* - \Delta i/2$: magnetisation (both switches ON).
- $i > i^* + \Delta i/2$: demagnetisation (both switches OFF).
- Otherwise: freewheeling (maintain previous state).

The switching frequency is variable, depending on DC-link voltage, inductance, and band width. A narrower band yields better current tracking but increases switching losses; 5–10% of rated current is a common design choice [3].

B. PWM Current Control

An alternative is fixed-frequency PWM applied to one switch while the other remains closed. The average phase voltage is $V_{\text{avg}} = D \cdot V_{dc}$, where $D \in [0, 1]$ is the duty cycle modulated by a PI current controller. PWM control offers a predictable switching frequency, simplifying EMI-filter design, but requires a current-feedback loop for effective regulation [2].

C. Turn-On / Turn-Off Angle Control

Above the base speed, the back-EMF prevents hysteresis or PWM controllers from maintaining the desired current, and the motor operates in *single-pulse* (voltage-control) mode. The primary control variables become:

- **Turn-on angle** θ_{on}: advancing it allows more time for current to build.
- **Turn-off angle** θ_{off}: advancing it reduces negative torque; delaying it increases average torque but risks torque reversal.

Optimal scheduling of $(\theta_{\text{on}}, \theta_{\text{off}})$ as a function of speed and load can be pre-computed and stored in a two-dimensional look-up table [1].

VI. Torque Ripple and Torque Sharing Functions

A. Sources of Ripple

Torque ripple is a principal SRM drawback, arising from: (i) discrete commutation between phases, creating torque dips; (ii) magnetic saturation limiting torque near the aligned position; and (iii) the inherently nonuniform $dL/d\theta$ profile. Reducing ripple is critical for applications requiring smooth speed regulation or low acoustic noise [5], [6].

B. TSF Formulations

TSF methods distribute the total torque reference T^* across overlapping phases so that their sum remains constant during commutation:

$$T_k^*(\theta) = \big(1 - f_r(\theta)\big)T^*, \quad T_{k+1}^*(\theta) = f_r(\theta)\, T^*, \tag{5}$$

where $f_r(\theta)$ is the rising sharing function defined over the overlap angle $[\theta_{\text{on}},\ \theta_{\text{on}} + \theta_{\text{ov}}]$. Table II compares common formulations.

TABLE II
Common Torque Sharing Function Formulations

TSF type	$f_r(\theta)$	Key trade-off
Linear	$(\theta - \theta_{\text{on}})/\theta_{\text{ov}}$	Simple; large $di/d\theta$ at edges
Sinusoidal	$\frac{1}{2}(1 - \cos\frac{\pi\theta}{\theta_{\text{ov}}})$	Smooth; moderate current stress
Cubic	$3x^2 - 2x^3,\ x = \theta/\theta_{\text{ov}}$	Zero slope at boundaries
Exponential	$1 - e^{-\alpha\theta}$	Tracks saturation; needs tuning

The feasibility of a TSF is ultimately limited by the available DC-link voltage: if the demanded di/dt exceeds $V_{dc}/L(\theta)$, actual torque deviates from reference and ripple increases [6], [7]. At high speeds, even smooth TSFs become infeasible, and advanced methods such as optimal current profiling [5]

or model-predictive control are employed. Individual phase torque references are inverted through a nonlinear $T(\theta, i)$ look-up table to obtain corresponding current references i_k^*, tracked by the inner current loop.

VII. Cascaded Speed Control Architecture

A standard outer-loop PI controller generates the total torque demand:

$$T^*(s) = K_p\, e_\omega(s) + \frac{K_i}{s}\, e_\omega(s), \tag{6}$$

with speed error $e_\omega = \omega^* - \omega$. An anti-windup mechanism is essential because the torque reference is saturated at a physical limit $T_{\max}$; without it, integrator windup causes significant overshoot during transient loading.

The TSF block distributes T^* among phases, the torque-to-current inversion obtains phase current references, and the inner current loop (hysteresis or PWM) tracks them via the converter. Position feedback—from an encoder or sensorless observer—is required at every level: the commutation logic, TSF indexing, and torque-to-current inversion all depend on θ [8]. This tight coupling between position knowledge and control performance underscores why accurate and fast position sensing is indispensable for SRM drives.

Gain tuning is typically performed via simulation-based trial-and-error or linearised small-signal models. A bandwidth separation of at least one decade between the inner current loop and the outer speed loop is recommended to preserve stability [2], [4].

VIII. Discussion and Next Milestone

The available project evidence is documentation-centric and does not yet include `.m` scripts, Simulink models, or waveform outputs. This paper therefore limits claims to theoretical readiness. The next milestone should provide: (1) a nonlinear SRM model implementation in MATLAB/Simulink, (2) baseline hysteresis and PWM controllers, (3) a TSF comparison under common operating points, and (4) closed-loop speed transient metrics including rise time, overshoot, and steady-state error.

IX. Conclusion

A complete theory baseline for SRM speed and torque control has been consolidated from project material and canonical references. The paper has detailed the machine construction and inductance characteristics, derived the governing electrical and mechanical equations, catalogued converter switching states, surveyed three current-control strategies (hysteresis, PWM, angle control), and presented TSF-based torque-ripple mitigation together with a cascaded PI speed-control architecture. These results provide a rigorous and traceable starting point for model-based implementation in subsequent milestones.

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